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Tray for display panel

Abstract

A tray for a display panel includes a body part including a base and an outer wall disposed to surround the base, and a slot part including a support member extending in a first direction and a second direction intersecting the first direction, and at least one buffer member disposed on an inner lateral side of the support member, wherein the at least one buffer member includes a plurality of buffer portions, each including airgaps, and a plurality of pillar portions, each disposed between adjacent ones of the plurality of buffer portions, the at least one buffer member includes a first surface, which is in contact with the support member, and a second surface, which faces the first surface, and the second surface has a convex shape in a plan view.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S)

(1) This application claims priority to and benefits of Korean Patent Application No. 10-2022-0086839 under 35 U.S.C. 119, filed on Jul. 14, 2022, in the Korean Intellectual Property Office (KIPO), the entire contents of which are incorporated herein by reference.

BACKGROUND

1. Technical Field

(2) The disclosure relates to a tray for a display panel.

2. Description of the Related Art

(3) Display devices have increasingly become of importance with the development of multimedia, and various types of display devices, such as a liquid crystal display (LCD) device, an organic light-emitting diode (OLED) display device, or the like, have been used.
(4) The LCD device includes an LCD panel, which displays an image using the light transmittance of liquid crystal molecules, and the OLED display device includes an OLED display panel.
(5) The LCD panel may include a polarizing plate thereon, and the OLED display panel may include an anti-reflective film thereon. The polarizing plate or the anti-reflective film may be attached to a lower substrate via an adhesive material, in which case, the LCD panel or the OLED display panel may be readily peeled off or damaged by external shock while being stored or transferred.
(6) To prevent damage to a display panel, the display panel may be carried around in a tray where storage space is formed. Multiple display panels may be stored and carried around in a state of being stacked in the tray.
(7) A tray for a display panel is formed of a Styrofoam material such as expanded polystyrene (EPS) or expanded polypropylene (EPP). However, a Styrofoam tray for a display panel has a relatively low impact resistance. Thus, multiple slots may be disposed in a tray for a display panel to correspond to the corners of a display panel. The slots can minimize the mobility of a display panel in the tray for a display panel and can thus prevent the display panel from being damaged by external shock.

SUMMARY

(8) Aspects of the disclosure provide a tray for a display panel, which is capable of minimizing external shock applied to a display panel.
(9) Aspects of the disclosure also provide a tray for a display panel, which is capable of minimizing the abrasion of slots.
(10) However, aspects of the disclosure are not restricted to those set forth herein. The above and other aspects of the disclosure will become more apparent to one of ordinary skill in the art to which the disclosure pertains by referencing the detailed description of the disclosure given below.
(11) According to an aspect of the disclosure, a tray for a display panel may include a body part including a base and an outer wall disposed to surround the base, and a slot part including a support member extending in a first direction and a second direction intersecting the first direction, and at least one buffer member disposed on an inner lateral side of the support member. The at least one buffer member may include a plurality of buffer portions, each including airgaps, and a plurality of pillar portions, each disposed between adjacent ones of the plurality of buffer portions. The at least one buffer member may include a first surface, which is in contact with the support member, and a second surface, which faces the first surface. The second surface may have a convex shape in a plan view.
(12) In an embodiment, a portion the at least one buffer member overlapping the plurality of pillar portion in the first direction or the second direction may not directly contact a display panel.
(13) In an embodiment, the at least one buffer member may further include a third surface, which is different from the first and second surfaces, and a fourth surface, which faces the third surface, and the airgaps may be formed of through holes extending from the third surface to the fourth surface.
(14) In an embodiment, the at least one buffer member may further include a third surface, which is different from the first and second surfaces, and a fourth surface, which faces the third surface, and the airgaps may be formed of pocket-type airgaps extending in a direction from the third surface to the fourth surface, penetrating the fourth surface, and not penetrating the third surface.
(15) In an embodiment, the support member may include a first support portion extending in the first direction and a second support portion extending in the second direction and intersecting the first support portion, and the at least one buffer member may include a first buffer member disposed on an inner lateral side of the first support portion and a second buffer member disposed on an inner lateral side of the second support portion.

(16) In an embodiment, an edge of the first buffer member and an edge of the second buffer member may contact each other.
(17) In an embodiment, an end of the first buffer member may protrude beyond an end of the first support portion.
(18) In an embodiment, an end of the first support portion may protrude beyond an end of the first buffer member.
(19) According to an embodiment of the disclosure, the tray for a display panel may include a body part including a base and an outer wall disposed to surround the base, and a slot part including a support member extending in a first direction and a second direction intersecting the first direction, and a buffer member disposed on an inner lateral side of the support member. The buffer member may include a plurality of buffer portions, each including airgaps, and a plurality of pillar portions, each disposed between adjacent ones of the plurality of buffer portions. Each of the plurality of buffer portions and the plurality of pillar portions may extend in the first or second direction.
(20) In an embodiment, each of the plurality of pillar portions may be arranged to be offset from a display panel in a lateral direction of the display panel.
(21) In an embodiment, the buffer member may be integrally bonded to the support member.
(22) In an embodiment, the buffer member may include a first surface and a second surface, which are disposed perpendicular to a direction the airgaps extend, and the airgaps may be pocket-type airgaps penetrating the first surface, and not penetrating the second surface.
(23) In an embodiment, the outer wall may include first coupling grooves disposed on inner lateral sides of the outer wall, the base may include second coupling grooves disposed on a top surface of the base, the support member may include support portions and coupling portions, the first coupling grooves may be coupled to the coupling portions, and the second coupling grooves may be coupled to the support portions and the buffer member.
(24) In an embodiment, the first coupling grooves and the second coupling grooves may be deeper than the top surface of the base in a third direction intersecting the first and second directions.
(25) According to an embodiment of the disclosure, a tray for a display panel may include a body part including a base and an outer wall disposed to surround the base, and a slot part including a support member extending in a first direction and a second direction intersecting the first direction, and a buffer member disposed on an inner lateral side of the support member. The buffer member may include a plurality of buffer portions, each including airgaps, and a plurality of pillar portions, each disposed between adjacent ones of the plurality of buffer portions. Each of the plurality of buffer portions and the plurality of pillar portions may extend in a third direction intersecting the first and second directions, and the buffer member may be integrally bonded to the support member.
(26) In an embodiment, the buffer member may include a first surface, and an entire area of the first surface may contact the support member.
(27) In an embodiment, a hardness of the support member may be greater than a hardness of the buffer member.
(28) In an embodiment, a Shore-A hardness of the buffer member may be in a range of about 80 to about 95.
(29) In an embodiment, the support member may include polycarbonate, and the buffer member may include urethane.
(30) In an embodiment, the body part may further include a receiving area that receives a display panel, and the receiving area may be disposed on the base inside of inner lateral sides of the outer wall in a plan view.
(31) According to the aforementioned and other embodiments of the disclosure, external shock applied to a display panel may be minimized.
(32) Also, the abrasion of slots may be minimized.

(33) It should be noted that the effects of the disclosure are not limited to those described above, and other effects of the disclosure will be apparent from the following description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) The above and other aspects and features of the disclosure will become more apparent by describing in detail embodiments thereof with reference to the attached drawings, in which:
(2) FIG. **1** is a perspective view of a tray for a display panel according to an embodiment of the disclosure;
(3) FIG. **2** is an exploded perspective view of the tray for a display panel according to an embodiment of the disclosure;
(4) FIG. **3** is a schematic cross-sectional view taken along line I-I′ of FIG. **2**;
(5) FIG. **4** is a plan view of the tray for a display panel according to an embodiment of the disclosure;
(6) FIG. **5** is an enlarged plan view of a slot part according to an embodiment of the disclosure;
(7) FIG. **6** is an enlarged perspective view of the slot part of FIG. **5**;
(8) FIG. **7** is an exploded perspective view of the slot part of FIG. **5** for explaining lateral sides of each buffer member;
(9) FIG. **8** is a schematic cross-sectional perspective view of the slot part of FIG. **6** taken along line II-II′;
(10) FIG. **9** is a schematic cross-sectional view taken along line II-II′ of FIG. **6**;
(11) FIG. **10** is a schematic cross-sectional perspective view of the slot part of FIG. **6** taken along line III-III′;
(12) FIG. **11** is a schematic cross-sectional perspective view of the slot part of FIG. **6** taken along line IV-IV′;
(13) FIG. **12** is a bottom view of the slot part of FIG. **5**;
(14) FIG. **13** is a plan view for explaining the buffer function of the slot part of FIG. **5**;
(15) FIGS. **14**A and **14**B show graphs for explaining the buffer function of the slot part of FIG. **5**;
(16) FIG. **15** is an enlarged perspective view of a slot part according to another embodiment of the disclosure;
(17) FIG. **16** is a schematic cross-sectional perspective view of the slot part of FIG. **15** taken along line V-V′;
(18) FIG. **17** is a schematic cross-sectional perspective view of the slot part of FIG. **15** taken along line VI-VI′;
(19) FIG. **18** is a schematic cross-sectional view of a tray for a display panel according to another embodiment of the disclosure;
(20) FIG. **19** is a graph for explaining the buffer function of the slot part of FIG. **15**;
(21) FIG. **20** is an enlarged perspective view of a slot part according to another embodiment of the disclosure;
(22) FIG. **21** is a schematic cross-sectional perspective view of the slot part of FIG. **20** taken along line VII-VII′;
(23) FIG. **22** is a plan view of a slot part according to another embodiment of the disclosure;
(24) FIG. **23** is a plan view for explaining the buffer function of the slot part of FIG. **22**;
(25) FIG. **24** is a perspective view of a slot part according to another embodiment of the disclosure;
(26) FIG. **25** is a plan view of the slot part of FIG. **24**;
(27) FIG. **26** is a perspective view of a slot part according to another embodiment of the disclosure;
(28) FIG. **27** is a plan view of the slot part of FIG. **26**;
(29) FIG. **28** is a perspective view of a slot part according to another embodiment of the disclosure;

and
(30) FIG. **29** is a plan view of the slot part of FIG. **28**.
DETAILED DESCRIPTION OF THE EMBODIMENTS
(31) The disclosure will now be described more fully hereinafter with reference to the accompanying drawings, in which embodiments of the disclosure are shown. This disclosure may, however, be embodied in different forms and should not be construed as limited to the embodiments set forth herein. Rather, these embodiments are provided so that this disclosure will be thorough and complete, and will fully convey the scope of the disclosure to those skilled in the art.
(32) When an element, such as a layer, is referred to as being “on”, “connected to”, or “coupled to” another element or layer, it may be directly on, connected to, or coupled to the other element or layer or intervening elements or layers may be present. When, however, an element or layer is referred to as being “directly on”, “directly connected to”, or “directly coupled to” another element or layer, there are no intervening elements or layers present. To this end, the term “connected” may refer to physical, electrical, and/or fluid connection, with or without intervening elements. Also, when an element is referred to as being “in contact” or “contacted” or the like to another element, the element may be in “electrical contact” or in “physical contact” with another element; or in “indirect contact” or in “direct contact” with another element. The same reference numbers indicate the same components throughout the specification.
(33) Unless otherwise defined or implied herein, all terms (including technical and scientific terms) used have the same meaning as commonly understood by those skilled in the art to which this disclosure pertains. It will be further understood that terms, such as those defined in commonly used dictionaries, should be interpreted as having a meaning that is consistent with their meaning in the context of the relevant art and should not be interpreted in an ideal or excessively formal sense unless clearly defined in the specification.
(34) Embodiments of the disclosure will hereinafter be described with reference to the accompanying drawings.
(35) FIG. **1** is a perspective view of a tray for a display panel according to an embodiment of the disclosure. FIG. **2** is an exploded perspective view of the tray for a display panel according to an embodiment of the disclosure. FIG. **3** is a schematic cross-sectional view taken along line I-I′ of FIG. **2**. FIG. **4** is a plan view of the tray for a display panel according to an embodiment of the disclosure.
(36) Referring to FIGS. **1** through **4**, first and second directions D**1** and D**2** may intersect each other. For example, the first and second directions D**1** and D**2** may perpendicularly intersect each other. A third direction D**3** may be, for example, a direction perpendicularly intersecting the first and second directions D**1** and D**2**.
(37) A tray **1000** for a display panel may receive a display panel DP therein for the purpose of transporting and storing the display panel DP.
(38) The display panel DP may be one of a liquid crystal display (LCD) panel, an electrophoretic display panel, a microelectromechanical system (MEMS) display panel, an electrowetting display panel, an organic light-emitting display panel, a micro-light-emitting diode (microLED) display panel, a quantum-dot display panel, and a quantum-rod display panel, but the type of the display panel DP is not particularly limited.
(39) The display panel DP may include a lower substrate **1**, an upper substrate **2**, an optical member **3**, and a seal member **4**. The lower substrate **1** may include switching elements such as thin-film transistors (TFTs) and other members for driving the switching elements. The upper substrate **2** may include members for display colors on the display panel DP, such as a color conversion layer and a color filter layer. The optical member **3** may be a polarizing plate in case that the display panel is an LCD panel or may be an anti-reflective film in case that the display panel DP is an organic light-emitting display panel.

(40) Although not specifically illustrated, the display panel DP may further include a window member, a driving member, a protective member, a backlight member, and a chassis. The window member may include a window glass and a window base layer. The driving member may include a circuit board (e.g., a driving circuit board) for driving the display panel DP. The protective member may include a protective film, a heat dissipation film, and a metal plate. The display panel DP may further include a light emitting device below the display panel DP if the display panel DP is an LCD panel. The light emitting device may include a light source and a light guide plate.
(41) The shape of the display panel DP is not particularly limited. The display panel DP is illustrated as a flat display panel, but the disclosure is not limited thereto. In another embodiment, the display panel DP may be a curved display panel.
(42) The tray **1000** may include a body part **100** and multiple slot parts **200**. The body part **100** may include a receiving area RA in which the display panel DP may be disposed. The slot parts **200** may be disposed to face the lateral sides of the display panel DP in the receiving area RA and may thus prevent the display panel DP from being damaged by external shock.
(43) The body part **100** may include a base **120** and an outer wall **110**, which is disposed along the outer circumference of the base **120**.
(44) The base **120** may provide a space in which the display panel DP may be disposed. The shape of the base **120** may be the same as, or conform to, the shape of the display panel DP.
(45) The outer wall **110** may be disposed along the circumference of the base **120**. The outer wall **110** may be disposed to surround the base **120**, in a plan view. The outer wall **110** may have a predetermined (or selectable) height and thickness.
(46) The outer lateral sides of the outer wall **110** may have a rectangular shape in a plan view. For example, the outer lateral sides of the outer wall **110** may have a rectangular shape having long sides extending in the first direction D**1** and short sides extending in the second direction D**2**. The corners at which the long sides and the short sides of each of the outer lateral sides of the outer wall **110** meet may be right-angled or may be rounded with a predetermined (or selectable) curvature. However, the shape of the outer lateral sides of the outer wall **110** is not particularly limited, and the outer lateral sides of the outer wall **110** may have various other shapes such as another polygonal shape, a circular shape, or an elliptical shape. The shape of the inner lateral sides of the outer wall **110** may vary depending on the shape of the display panel DP. For example, the outer wall **110** may include receiving grooves, which are inwardly recessed to receive the circuit board of the display panel DP.
(47) The outer wall **110** may include first coupling grooves **111**. The first coupling grooves **111** may be disposed on the inner lateral sides of the outer wall **110**. The first coupling grooves **111** may be configured to be coupled to coupling portions **212** of support members **210** of the slot parts **200**.
(48) The base **120** may include second coupling grooves **121**. The second coupling grooves **121** may be disposed on the top surface of the base **120**. The second coupling grooves **121** may be configured to be coupled to support portions **211** of the support members **210** of the slot parts **200** and buffer members **220** of the slot parts **200**.
(49) As the body part **100** is provided with the first coupling grooves **111** and the second coupling grooves **121**, the slot parts **200** may be coupled to, or decoupled from, the body part **100** via the first coupling grooves **111** and the second coupling grooves **121**. As the slot parts **200** can be freely coupled to, or decoupled from, the tray **1000** via the first coupling grooves **111** and the second coupling grooves **121**, without regard to the size and the shape of the tray **1000**, the slot parts **200** may be readily replaced and used.
(50) The thickness and the height of the first coupling grooves **111** and the thickness and the height of the coupling portions **212** of the support members **210** of the slot parts **200** may be substantially the same. The thickness of the second coupling grooves **121** and the sum of the thickness of the support members **210** of the slot parts **200** and the thickness of the buffer members **220** of the slot parts **200** may be substantially the same. For example, the first coupling grooves **111** and the

second coupling grooves **121** may be formed deeper than the top surface of the base **120** in the third direction D**3**. For example, the first coupling grooves **111** and the second coupling grooves **121** may be formed to penetrate a plane extended in the first and second directions D**1** and D**2** from the top surface of the base **120**. As a result, the slot parts **200** may be firmly coupled to the body part **100**.
(51) The outer wall **110** of the body part **100** may define the receiving area RA where the display panel DP can be received, together with the base **120**. The receiving area RA may be positioned in the space surrounded by the outer wall **110**, on the top surface of the base **120**. The receiving area RA may have the same area as, or a larger area than, the display panel DP, which is received in the receiving area RA. The shape of the receiving area RA may vary depending on the shape of the display panel DP.
(52) The boundaries (or edges) of the receiving area RA may be positioned inside of the outer wall **110** in a plan view. For example, the boundaries of the receiving area RA may coincide with the inner boundaries of the buffer members **220** of the slot parts **200** or may be positioned inside of the inner boundaries of the buffer members **220** of the slot parts **200** in a plan view. The lateral sides of the display panel DP may only contact the slot parts **200**, but not the outer wall **110**. Thus, the display panel DP may be prevented from colliding with the outer wall **110** due to external shock, and the stress applied to the display panel DP may be minimized by the buffer members **220**.
(53) In some embodiments, the body part **100** may be formed of Expanded polystyrene (EPS), Expanded polypropylene (EPP), or a mixture thereof. In case that the body part **100** is formed of a Styrofoam material such as EPS or EPP, the body part **100** may have a relatively low impact resistance. However, as the tray **1000** includes the slot parts **200**, which have a shock absorbing function, external shock may be prevented from being directly delivered to the display panel DP through the body part **100**, which has a relatively low impact resistance.
(54) The slot parts **200** may be disposed on the inner lateral sides of the outer wall **110** of the body part **100**. The slot parts **200** may be disposed near the corners of the body part **100**. The number and the locations of the slot parts **200** are not particularly limited. For example, four slot parts **200** may be disposed at the corners of the tray **1000** to contact the inner lateral sides of the outer wall **110**.
(55) The slot parts **200** may be coupled to the body part **100** through the first coupling grooves **111** and the second coupling grooves **121**. The coupling portions **212** of the support members **210** of the slot parts **200** may be coupled to the first coupling grooves **111** of the outer wall **110**, and the support members **210** and the buffer members **220** of the slot parts **200** may be coupled to the second coupling grooves **121** of the base **120**. The slot parts **200** may be fitted and fixed to the body part **100** through the first coupling grooves **111** and the second coupling grooves **121**.
(56) The slot parts **200** will hereinafter be described with reference to FIGS. **5** through **12**.
(57) FIG. **5** is an enlarged plan view of a slot part according to an embodiment of the disclosure. FIG. **6** is an enlarged perspective view of the slot part of FIG. **5**. FIG. **7** is an exploded perspective view of the slot part of FIG. **5** for explaining lateral sides of each buffer member. FIG. **8** is a schematic cross-sectional perspective view of the slot part of FIG. **6** taken along line II-II′. FIG. **9** is a schematic cross-sectional view taken along line II-II′ of FIG. **6**. FIG. **10** is a schematic cross-sectional perspective view of the slot part of FIG. **6** taken along line III-III′. FIG. **11** is a schematic cross-sectional perspective view of the slot part of FIG. **6** taken along line IV-IV′. FIG. **12** is a bottom view of the slot part of FIG. **5**.
(58) Referring to FIGS. **5** through **12**, a slot part **200** may include a support member **210** and one or more buffer members **220**, which are disposed on the support member **210**.
(59) The support member **210** may include a support portion **211** and a coupling portion **212**, which are L-shaped. The support portion **211** may include first and second support portions **211***a* and **211***b*, which extend in the first and second directions D**1** and D**2**, respectively. The support portion **211** may be configured such that the buffer members **220** may be bonded to a surface of the support portion **211**. The coupling portion **212** may include first and second coupling portions **212***a* and

212*b*, which extend in the first and second directions D**1** and D**2**, respectively. The coupling portion **212** may be coupled to the first coupling groove **111** of the outer wall **110**. The thickness and the height of the coupling portion **212** and the thickness and the height of the first coupling groove **111** may be substantially the same. Thus, the slot part **200** may be stably fixed to the body part **100**.

(60) The buffer members **220** may control the mobility of the display panel DP and absorb and relieve impact that may be generated upon the movement of the display panel DP. The buffer members **220** may be disposed directly on the inner lateral sides of the support portion **211** of the support member **210**. Each of the buffer members **220** may include a first buffer portion, which is disposed on the inner lateral side of the first support portion **211***a*, and a second buffer portion, which is disposed on the inner lateral side of the second support portion **211***b*.

(61) In some embodiments, the buffer members **220** may be bonded to the support portion **211** of the support member **210**. The buffer members **220** is illustrated as having a cuboid shape in FIG. **5**-**8**, but the shape of the buffer members **220** is not particularly limited.

(62) Each of the buffer members **220** may include multiple buffer portions **221**, multiple pillar portions **222**, a bonding portion **223**, and a panel contact portion **224**. The buffer portions **221** may be disposed between the bonding portion **223** and the panel contact portion **224**. The buffer portions **221** may include airgaps, which are in the form of through holes or pockets. The pillar portions **222** may be disposed between the buffer portions **221** to connect the bonding portion **223** and the panel contact portion **224**. The bonding portion **223** may be disposed on, and in direct contact with, the support portion **211** of the support member **210**. The buffer member **220** may be bonded to the support member **210** through the bonding portion **223**. The panel contact portion **224** may be disposed on the opposite side of the bonding portion **223** with respect to the buffer portions **221** and the pillar portions **222**, which are interposed between the panel contact portion **224** and the bonding portion **223**. The panel contact portion **224** may be configured to be in direct contact with the display panel DP.

(63) Referring to FIG. **7**, each of the buffer members **220** may have first through sixth surfaces **220***a* through **220***f*. The first surface **220***a* may be in direct contact with the support member **210**, the second surface **220***b* may be disposed opposite to the first surface **220***a*, the third surface **220***c* may face the base **120**, the fourth surface **220***d* may be disposed opposite to the third surface **220***c*, the fifth surface **220***e* may face the support member **210** without being in direct contact with the support member **210**, and the sixth surface **220***f* may be disposed opposite to the fifth surface **220***e*.

(64) The buffer portions **221** of each of the buffer members **220** may include airgaps. The airgaps of the buffer portions **221** may extend in the third direction D**3**. The airgaps of the buffer portions **221** may be formed as through holes penetrating the third and fourth surfaces **220***c* and **220***d*. For example, two buffer members **220** may be provided, and three airgaps may be formed in each of the buffer portions **221** of the buffer members **220**. However, the number of airgaps formed in each of the buffer members **220** is not particularly limited. Also, cuboid airgaps are illustrated as being formed in each of the buffer members **220**, but the shape of the airgaps of the buffer portions **221** of each of the buffer members **220** may vary. In another embodiment, cylindrical airgaps or airgaps in various other shapes may be formed in each of the buffer members **220**.

(65) In each of the buffer members **220**, the buffer portions **221** may be completely surrounded by the pillar portions **222**, the bonding portion **223**, and the panel contact portion **224**, in a plan view (as viewed in the third direction D**3**). For example, referring to FIG. **6**, in each of the buffer members **220**, the buffer portions **221** may be completely surrounded by the pillar portions **222**, the bonding portion **223**, and the panel contact portion **224**, as viewed in the third direction D**3**, i.e., as viewed from above each of the buffer members **220**.

(66) In each of the buffer members **220**, the airgaps of the buffer portions **221** may be disposed between the bonding portion **223** and the panel contact portion **224** in the first direction D**1**, in a cross-sectional view taken along the plane defined by the first and third directions D**1** and D**3**. In

another embodiment, in each of the buffer members **220**, the airgaps of the buffer portions **221** may be disposed between the bonding portion **223** and the panel contact portion **224** in the second direction D**2**, in a cross-sectional view taken along the plane defined by the second and third directions D**2** and D**3**. For example, referring to FIG. **8**, in each of the buffer members **220**, the airgaps of the buffer portions **221** may be disposed between the bonding portion **223** and the panel contact portion **224** in the first direction D**1**, in a cross-sectional view taken along line II-II′ of FIG. **6**. In a cross-sectional view taken along line II-II′ of FIG. **6**, the buffer portions **221** of each of the buffer members **220** may be spaced apart from the support member **210** by the bonding portions **223** of the buffer members **220**.

(67) The pillar portions **222** may be disposed between adjacent buffer portions **221** in the first or second direction D**1** or D**2**. The pillar portions **222** may extend in the third direction D**3**.

(68) In each of the buffer members **220**, the pillar portions **222** may be surrounded by the buffer portions **221**, the bonding portion **223**, and the panel contact portion **224**. For example, referring to FIG. **6**, in each of the buffer members **220**, the pillar portions **222** may be completely surrounded by the buffer portions **221**, the bonding portion **223**, and the panel contact portion **224**, as viewed in the third direction D**3**, i.e., as viewed from above each of the buffer members **220**.

(69) In each of the buffer members **220**, the pillar portions **222** may be disposed between the bonding portion **223** and the panel contact portion **224** in the first direction D**1**, in a cross-sectional view taken along the plane defined by the first and third directions D**1** and D**3**. In another embodiment, in each of the buffer members **220**, the pillar portions **222** may be disposed between the bonding portion **223** and the panel contact portion **224** in the second direction D**2**, in a cross-sectional view taken along the plane defined by the second and third directions D**2** and D**3**. For example, referring to FIG. **8**, in each of the buffer members **220**, the pillar portions **222** may be disposed between the bonding portion **223** and the panel contact portion **224** in the first direction D**1**, in a cross-sectional view taken along line II-II′ of FIG. **6**. In a cross-sectional view taken along line II-II′ of FIG. **6**, the pillar portions **222** of each of the buffer members **220** may be spaced apart from the support member **210** by the bonding portions **223** of the buffer members **220**.

(70) In each of the buffer members **220**, the bonding portion **223** and the panel contact portion **224** may extend in the first and third directions D**1** and D**3**. In another embodiment, in each of the buffer members **220**, the bonding portion **223** and the panel contact portion **224** may extend in the second and third directions D**2** and D**3**. The bonding portions **223** of the buffer members **220** may be disposed adjacent to the support member **210**, and the panel contact portions **224** of the buffer members **220** may be disposed on the opposite sides of the bonding portions **223** of the buffer members **220**. A lateral side (e.g., the first surfaces **220***a*) of the bonding portions **223** of the buffer members **220** may be in direct contact with the support member **210**. As a result, as the bonding area between the buffer members **220** and the support member **210** increases, the buffer members **220** may be firmly bonded to the support member **210**.

(71) Referring to FIG. **9**, a thickness d**3** of the panel contact portions **224** of the buffer members **220** may be less than a thickness d**1** of the bonding portions **223** of the buffer members **220**, but embodiments are not limited thereto. For example, the thickness d**1** of the bonding portions **223** of the buffer members **220** may be in a range of about 0.5 mm to about 1.5 mm, and the thickness d**3** of the panel contact portions **224** of the buffer members **220** may be in a range of about 1.0 mm to about 2.0 mm.

(72) A thickness d**2** of the airgaps of the buffer portions **221** of the buffer members **220** may be greater than the thicknesses d**1** and d**3**. For example, the thickness d**2** of the airgaps of the buffer portions **221** of the buffer members **220** may be in a range of about 1.5 mm to about 2.5 mm. As a result, as the airgaps of the buffer portions **221** of the buffer members **220** includes more air, the buffering effect of the buffer members **220** may be improved.

(73) The slot part **200** may be formed by a mold manufacturing process. The support member **210** may be formed by injecting a material for forming the support member **210** into a mold, and the

buffer members **220** may be formed by injecting a material for forming the buffer members **220** into the mold. During the formation of the support member **210** and the buffer members **220**, pressure may be applied, or processes such as heat treatment may be performed. As already mentioned above, the support member **210** and the buffer members **220** may be bonded to each other and may thus be formed in one body. For example, the support member **210** and the buffer members **220** may be integrally formed by injection. Also, as already mentioned above, as the entire lateral side of the buffer members **220** is bonded to the support member **210**, the bonding area of the buffer members **220** and the support member **210** increases, and thus, the buffer members **220** may be firmly bonded to the support member **210**. Accordingly, no gaps may be included over the entire bonding surface between the support member **210** and each of the buffer members **220**. As the support member **210** and the buffer members **220** are firmly bonded together with almost zero gaps therebetween, the mobility of the display panel DP may be effectively controlled.

(74) In some embodiments, a synthetic resin having an excellent impact resistance and capable of being formed by injection molding or thermal molding, such as polycarbonate (PC), may be used as the material of the support member **210**, and a material capable of performing a buffer function, such as urethane, may be used as the material of the buffer members **220**.

(75) The hardness of the support member **210** may be greater than the hardness of the buffer members **220**. The hardness of a support member **210** formed of polycarbonate may be greater than the hardness of buffer members **220** formed of urethane. For example, the hardness of the support member **210** may be in a range of about 90 to about 100, and the hardness of the buffer members **220** may be in a range of about 70 to about 95. The term “hardness,” as used herein, may be a Shore A hardness measurement obtained using a Shore A-type indenter. The support member **210** may be formed of a material that is relatively harder than the material of the buffer members **220** to prevent the mobility of the display panel DP and protect the display panel DP from external impact. On the contrary, the buffer member **220** may be formed of a material that is softer than the material of the support member **210** to reduce stress that may be caused upon direct contact with the display panel DP.

(76) In some embodiments, the buffer members **220** may have a Shore-A hardness of in a range of about 80 to about 95. In case that the Shore-A hardness of the buffer members **220** is less than about 80, the amount of impact applied to the display panel DP may be reduced, and the stress applied to the display panel DP may be lowered. However, the buffer members **220** may be worn away due to their relatively low hardness, and as a result, a gap may be formed between the display panel DP and the slot part **200**, and the mobility of the display panel DP may increase. Accordingly, in case that the hardness of the buffer members **220** is greater than or equal to about 80, the abrasion of the buffer members **220** may be prevented.

(77) The buffer function of the slot part of FIG. **5** will hereinafter be described with reference to FIGS. **13** and **14**.

(78) FIG. **13** is a plan view for explaining the buffer function of the slot part of FIG. **5**, and FIGS. **14**A and **14**B show graphs for explaining the buffer function of the slot part of FIG. **5**. FIG. **14**A is a graph showing the level of stress applied to a display panel upon collision with a conventional buffer member having no buffer portions and no pillar portions, and FIG. **14**B is a graph showing the level of stress applied to a display panel upon collision with a buffer member of the slot part of FIG. **5**, i.e., a buffer member **220** having buffer portions **221** and pillar portions **222**.

(79) Referring to FIGS. **13**, **14**A, and **14**B, the display panel DP may be in direct contact with the buffer member **220**. For example, lateral sides of the display panel DP may be in direct contact with a second surface **220***b* of the buffer member **220**. Thus, an impact may be transmitted to the buffer member **220** due to the mobility of the display panel DP. This impact may be transmitted to the display panel DP by a reaction, and as a result, stress may be applied to the display panel DP. As shown in FIG. **14**B, the amount of impact applied to the display panel DP by a reaction may be

reduced.
(80) For example, the buffer member **220** may include the buffer portions **221** and the pillar portions **222**. As the buffer portions **221** include airgaps, the amount of impact returned to the display panel DP by a reaction may be reduced. The pillar portions **221**, which have elasticity, may be bent in a direction perpendicular to a force generated by the display panel DP and may thus distribute the force. For example, the pillar portions **222** may be bent in a direction perpendicular to the second surface **220***b* of the buffer member **220** and may thus reduce the amount of impact returned to the display panel DP by a reaction.
(81) For example, as shown in FIG. **14**A, in case that the conventional buffer member having no buffer portions and no pillar portions has a hardness of 80 or greater, the optical member **3** above the display panel DP may be detached from the upper substrate **2** or the seal member **4**. On the contrary, as shown in FIG. **14**B even in case that the buffer member **220** has a hardness of 80 or greater, the stress applied to the display panel DP may be reduced due to the buffer portions **221** and the pillar portions **222**, and as a result, the detachment of the optical member **3** may be minimized or prevented.
(82) The slot part **200** according to an embodiment of the disclosure may reduce the amount of impact returned to the display panel DP by a reaction. Also, as the amount of impact returned to the display panel DP by a reaction is reduced, the hardness of the buffer member **220** may be set relatively high, and thus, the abrasion of the buffer member **220** may be prevented.
(83) Slot parts according to other embodiments of the disclosure will hereinafter be described, focusing on the differences with the slot part of FIG. **5**.
(84) Slot parts according to other embodiments of the disclosure will hereinafter be described with reference to FIGS. **15** through **19**.
(85) FIG. **15** is an enlarged perspective view of a slot part according to another embodiment of the disclosure. FIG. **16** is a schematic cross-sectional perspective view of the slot part of FIG. **15** taken along line V-V′. FIG. **17** is a schematic cross-sectional perspective view of the slot part of FIG. **15** taken along line VI-VI′. FIG. **18** is a schematic cross-sectional view of a tray for a display panel according to another embodiment of the disclosure. FIG. **19** is a graph for explaining the buffer function of the slot part according to another embodiment of the disclosure.
(86) Referring to FIGS. **15** through **19**, the slot part **200_1** is different from the slot part **200** of FIG. **6** in that the airgaps of the buffer portions **221_1** and the pillar portions **222_1** of each buffer member **220_1** extend in the first or second direction D**1** or D**2**.
(87) For example, the airgaps of the buffer portions **221_1** may extend in the first or second direction D**1** or D**2**. The airgaps of the buffer portions **221_1** may be through holes penetrating a fifth or sixth surface **220***e* or **220***f* of each buffer member **220_1**.
(88) The buffer portions **221_1** may be completely surrounded by the pillar portions **2221**, the bonding portion **223**, and the panel contact portion **224**, on a plane as viewed in the first or second direction D**1** or D**2**. For example, referring to FIG. **15**, the buffer portions **2211** may be surrounded by the pillar portions **2221**, the bonding portion **223**, and the panel contact portion **224**, on a plane as viewed from the front of each buffer member **220_1**.
(89) The airgaps of the buffer portions **2211** may be disposed between the bonding portion **223** and the panel contact portion **224**, in the first or second direction D**1** or D**2**, on a plane defined by the first and second directions D**1** and D**2**. For example, referring to FIG. **16**, the airgaps of the buffer portions **221_1** may be disposed between the bonding portion **223** and the panel contact portion **224** in the first direction D**1**. The buffer portions **221_1** may be spaced apart from the support member **210** by the bonding portion **223**, in a cross-sectional view taken along line V-V′ of FIG. **15**.
(90) The pillar portions **222_1** may be disposed between the buffer portions **221_1** in the third direction D**3**. The pillar portions **222_1** may extend in the first or second direction D**1** or D**2**.
(91) The pillar portions **222_1** may be completely surrounded by the buffer portions **2211**, the

bonding portion **223**, and the panel contact portion **224**, on the plane as viewed in the first or second direction D**1** or D**2**. For example, referring to FIG. **15**, the pillar portions **2221** may be completely surrounded by the buffer portions **2211**, the bonding portion **223**, and the panel contact portion **224**, as viewed in the first or second direction D**1** or D**2**, i.e., as viewed from the front or from a side of each buffer member **220_1**.
(92) The pillar portions **222_1** may be disposed between the bonding portion **223** and the panel contact portion **224** in the first or second direction D**1** or D**2**, in a cross-sectional view taken along the plane defined by the first and second directions D**1** and D**2**. The pillar portions **222_1** may be spaced apart from the support member **210** by the bonding portion **223**, in a cross-sectional view taken along the plane defined by the first and second directions D**1** and D**2**. For example, referring to FIG. **16**, the pillar portions **222_1** may be disposed between the bonding portion **223** and the panel contact portion **224** in the first direction D**1**. The pillar portions **222_1** may be spaced apart from the support member **210** by the bonding portion in the first direction D**1**.
(93) Referring to FIG. **18**, the pillar portions **222_1** of the buffer member **220_1** of the tray **1000_1** may be arranged to be offset from a display panel DP in the lateral direction of the display panel DP. Accordingly, as a force applied from the display panel DP is not transferred perpendicularly to the pillar portions **2221**, the stress applied to the display panel DP may be reduced, as shown in FIG. **19**.
(94) A slot part according to another embodiment of the disclosure will hereinafter be described with reference to FIGS. **20** and **21**.
(95) FIG. **20** is an enlarged perspective view of a slot part according to another embodiment of the disclosure. FIG. **21** is a schematic cross-sectional perspective view of the slot part of FIG. **20** taken along line VII-VII′.
(96) Referring to FIGS. **20** and **21**, the slot part **200_2** is different from the slot part **200** of FIG. **6** in that the airgaps of the buffer portions **221_2** of each buffer member **220_2** are formed as pockets, not as through holes.
(97) For example, the airgaps of the buffer portions **221_2** may be pocket-type airgaps penetrating the fourth surface **220***d* of each buffer member **220_2**, but not penetrating the third surface **220***c* of each buffer member **220_2**. First ends of the buffer portions **221_2** may be disposed on the fourth surface **220***d*, and second ends of the buffer portions **221_2** may be disposed in the buffer member **220_2**. The pillar portion **2222** may be disposed between the buffer portions **221_2**. As the airgaps of the buffer portions **221_2** are formed as pocket-type airgaps penetrating a surface, but not another surface, of each buffer member **220_2**, the structure of a mold for forming the slot part **200_2** may be simplified, and the amount of impact may be controlled by controlling the size of the airgaps of the buffer portions **221_2**.
(98) A slot part according to another embodiment of the disclosure will hereinafter be described with reference to FIGS. **22** and **23**.
(99) FIG. **22** is a plan view of a slot part according to another embodiment of the disclosure. FIG. **23** is a plan view for explaining the buffer function of the slot part of FIG. **22**.
(100) Referring to FIGS. **22** and **23**, the slot part **200_3** is different from the slot part **200** of FIG. **6** in that the panel contact portion **224_3** of each buffer member **220_3** has a convex shape.
(101) For example, the panel contact portion **224_3** of each buffer member **220_3** may have a convex shape in a plan view. For example, the second surface **220***b* of each buffer member **220_3** may not be flat, and may be convex.
(102) As illustrated in FIG. **23**, a display panel DP may contact a most protruding part of the second surface **220***b*, which is convex, and then portions adjacent to the most protruding part of the second surface **220***b*.
(103) In case that a force is transferred perpendicularly to the buffer member **2203**, the force may be distributed to the buffer member in other directions. Accordingly, the amount of impact returned to the display panel DP by a reaction may be reduced.

(104) By controlling the curvature of the second surface **220***b* of each buffer member **220_3** or the locations of pillar portions **222**, the display panel DP may not directly contact the buffer member **220_3** in areas overlapping the pillar portions **222** in the first or second direction D**1** or D**2**.
(105) As the display panel DP does not directly contact the buffer member **220_3** in the areas overlapping the pillar portions **222**, a force may not be perpendicularly applied by the display panel DP to the pillar portions **222**, and as a result, the stress applied to the display panel DP may be reduced.
(106) A slot part according to another embodiment of the disclosure will hereinafter be described with reference to FIGS. **24** and **25**.
(107) FIG. **24** is a perspective view of a slot part according to another embodiment of the disclosure. FIG. **25** is a plan view of the slot part of FIG. **24**.
(108) Referring to FIGS. **24** and **25**, the slot part **200_4** is different from the slot part **200** of FIG. **6** in that edges of the buffer members **220_4** contact with each other.
(109) For example, an edge of the buffer member **220_4** on the first support portion **211***a* and an edge of the buffer member **220_4** on the second support portion **211***b* may contact each other. Accordingly, as the areas of contact between the buffer member **220_4** and the first support portion **211***a* and between the buffer member **220_4** and the second support portion **211***b* increase, the first and second support portions **211***a* and **211***b* may be further firmly bonded to the buffer members **220_4**. Also, as the slot part **200_4** completely covers even the corners of a display panel DP, the mobility of the display panel DP may be further reduced.
(110) Slot parts according to other embodiments of the disclosure will hereinafter be described with reference to FIGS. **26** through **29**.
(111) FIG. **26** is a perspective view of a slot part according to another embodiment of the disclosure. FIG. **27** is a plan view of the slot part of FIG. **26**. FIG. **28** is a perspective view of a slot part according to another embodiment of the disclosure. FIG. **29** is a plan view of the slot part of FIG. **28**.
(112) Referring to FIGS. **26** and **27**, the slot part **200_5** is different from the slot part **200** of FIG. **6** in that the buffer members **2205** protrude beyond the support member **210** in the first or second direction D**1** or D**2**. Referring to FIGS. **28** and **29**, the slot part **200_6** is different from the slot part **200** of FIG. **6** in that the support member **210** protrude beyond the buffer members **220_6** in the first or second direction D**1** or D**2**.
(113) The above description is an example of technical features of the disclosure, and those skilled in the art to which the disclosure pertains will be able to make various modifications and variations. Therefore, the embodiments of the disclosure described above may be implemented separately or in combination with each other.
(114) Therefore, the embodiments disclosed in the disclosure are not intended to limit the technical spirit of the disclosure, but to describe the technical spirit of the disclosure, and the scope of the technical spirit of the disclosure is not limited by these embodiments.

Claims

1. A tray for a display panel, comprising: a body part including a base and an outer wall surrounding the base; and a slot part including a support member extending in a first direction and a second direction intersecting the first direction, and one or more buffer members disposed on an inner lateral side of the support member, wherein a buffer member of the one or more buffer members includes a plurality of buffer portions, each including a corresponding airgap, and a plurality of pillar portions, each disposed between adjacent buffer portions of the plurality of buffer portions, the buffer member includes: a first surface, which faces and is in contact with the inner lateral side of the support member; a second surface, which opposes the first surface; a third surface, which is different from the first and second surfaces; and a fourth surface, which opposes

the third surface, the second surface has a convex shape protruding laterally away from the inner lateral side of the support member, and the corresponding airgaps are formed of pocket-type airgaps extending in a direction from the third surface to the fourth surface, penetrating the fourth surface, and not penetrating the third surface.

2. The tray of claim 1, wherein, in a case that the display panel is supported by the tray at least via the buffer member, a portion of the buffer member overlapping the plurality of pillar portions in the first direction or the second direction does not directly contact the display panel.

3. The tray of claim 1, wherein the support member includes a first support portion extending in the first direction and a second support portion extending in the second direction and intersecting the first support portion, and the buffer member comprises a first buffer member disposed on an inner lateral side of the first support portion and a second buffer member disposed on an inner lateral side of the second support portion.

4. A tray for a display panel, comprising: a body part including a base and an outer wall surrounding the base; and a slot part including a support member extending in a first direction and a second direction intersecting the first direction, and a buffer member disposed on an inner lateral side of the support member, wherein the buffer member includes: a first surface, which faces and is in contact with the inner lateral side of the support member; a second surface, which opposes the first surface; a third surface, which is different from the first and second surfaces; a fourth surface, which opposes the third surface; a plurality of buffer portions, each including a corresponding airgap; and a plurality of pillar portions, each disposed between adjacent buffer portions of the plurality of buffer portions, each of the plurality of buffer portions and the plurality of pillar portions extends in a third direction intersecting the first and second directions, the corresponding airgaps are formed of pocket-type airgaps extending in a direction from the third surface to the fourth surface, penetrating the fourth surface, and not penetrating the third surface, and the buffer member is integrally bonded to the support member.

5. The tray of claim 4, wherein an entire area of the first surface contacts the support member.

6. The tray of claim 4, wherein a hardness of the support member is greater than a hardness of the buffer member.

7. The tray of claim 6, wherein a Shore-A hardness of the buffer member is in a range of about 80 to about 95.

8. The tray of claim 7, wherein the support member includes polycarbonate, and the buffer member includes urethane.

9. The tray of claim 4, wherein the body part further includes a receiving area that receives the display panel, and the receiving area is disposed on the base inside of inner lateral sides of the outer wall in a view in the third direction.